

t – test and friends

comparing means of two independent samples

non-normal data,
unequal variances,
outliers,
covariate adjustments

Goal	Problem to solve	Requirements	Method
Compare 2 means		Gaussian distr. + equal variances	t – test
	Unequal variances	Gaussian distr.	Welch t – test with Satterthwaite degrees of freedom
	Outliers	Gaussian distr. (without outliers) + equal variances	Yuen t – test or robust (MM-estim.) regression followed by Wald test Estimand change: means → trimmed / weighted means
	Outliers + unequal variances	Gaussian distributions	Yuen – Welch t or robust regr. + HC2/HC3 SE + Wald test Estimand change: as above
	Non-Gaussian distributions	Equal variances (exchangeability)	Permutation or bootstrapped t – test / ≈ Fisher-Pitman /
	Non-Gaussian distributions + unequal variances		Permutation Welch t – test (Janssen, 1997)
Compare 2 means adjusted for covariate(s)		Gaussian distr. + equal variances, parallel slopes	OLS – fit ANCOVA
	Unequal variances	Gaussian distr. + equal variances, parallel slopes	OLS – fit ANCOVA + HC (sandwich) SE GLS – fit ANCOVA
	Non-Gaussian distributions	Equal variances, parallel slopes	Freedman-Lane permutation ANCOVA
	Non-Gaussian distributions + unequal variances (+ high-leverage observations)	Parallel slopes	ANCOVA via wild bootstrap (Wu) with Rademacher or Mammen weights + (opt.) HC2 / HC3 SE